DESIGN AND MODELING OF A WATER BOTTLE USING AUTODESK FUSION 360

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1. Introduction

Computer-Aided Design (CAD) plays a vital role in modern product design and manufacturing. It allows engineers and designers to create precise, accurate, and realistic models of products before they are manufactured. CAD software helps in reducing errors, saving time, and improving product quality. One such powerful CAD tool is **Autodesk Fusion 360**, which integrates design, modeling, assembly, and visualization into a single platform.

In this project, a **water bottle and its cap** have been designed and assembled using Autodesk Fusion 360. Bottled water containers are widely used in daily life and require careful consideration of shape, strength, usability, and compatibility between the bottle and cap. This project focuses on creating a realistic 3D model of a bottle similar to a commercial water bottle, along with a properly fitting cap.

The project introduces fundamental concepts of **solid modeling, part design, text embossing, and assembly modeling**, making it suitable for beginners in CAD design as well as students learning product design.

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2. Project Objective

The main objectives of this project are:

- To design a **3D model of a water bottle** using Autodesk Fusion 360.
- To create a **separate cap model** compatible with the bottle neck.
- To assemble the bottle and cap accurately using assembly tools.
- To understand and apply **basic solid modeling operations** such as sketching, extrusion, filleting, and threading.
- To enhance skills in **visualization and presentation** of 3D models.
- To gain practical exposure to designing real-world consumer products.

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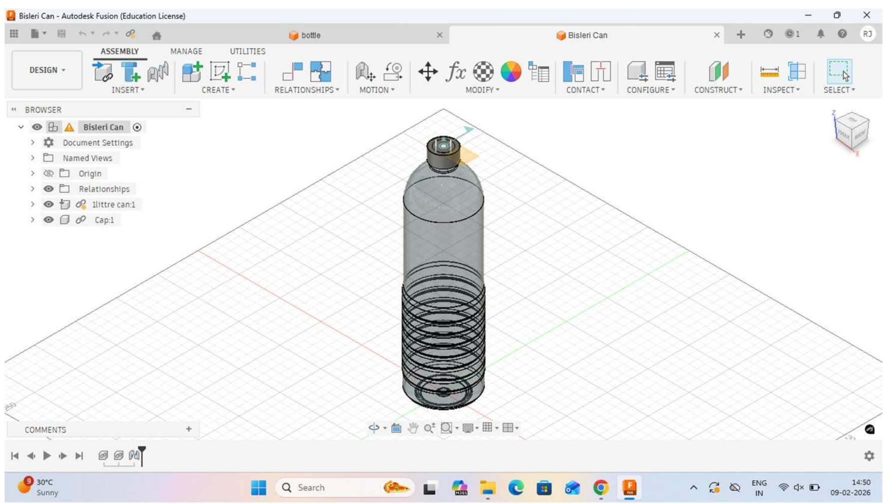
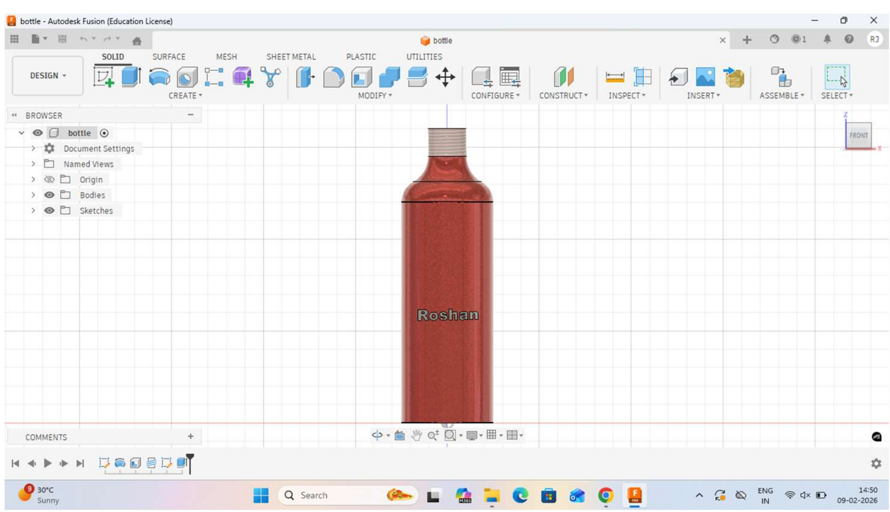
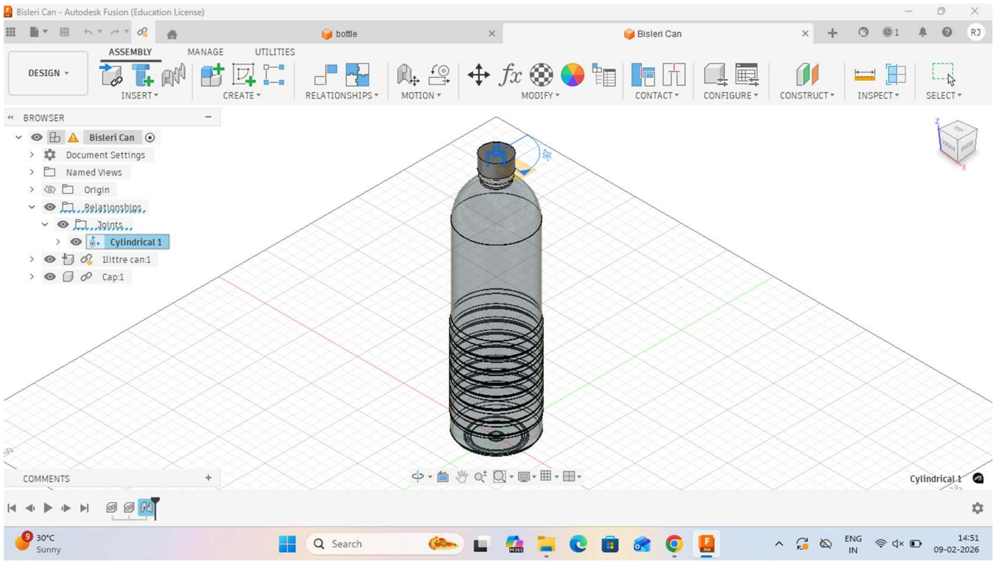
3. Software and Tools Used

- **Software:** Autodesk Fusion 360 (Education License)
- **Workspace Used:**
  - Solid Workspace
  - Assembly Workspace
- **Modeling Tools:**
  - Sketch
  - Extrude
  - Fillet
  - Text / Emboss
  - Appearance
  - Assembly Constraints

4.Images



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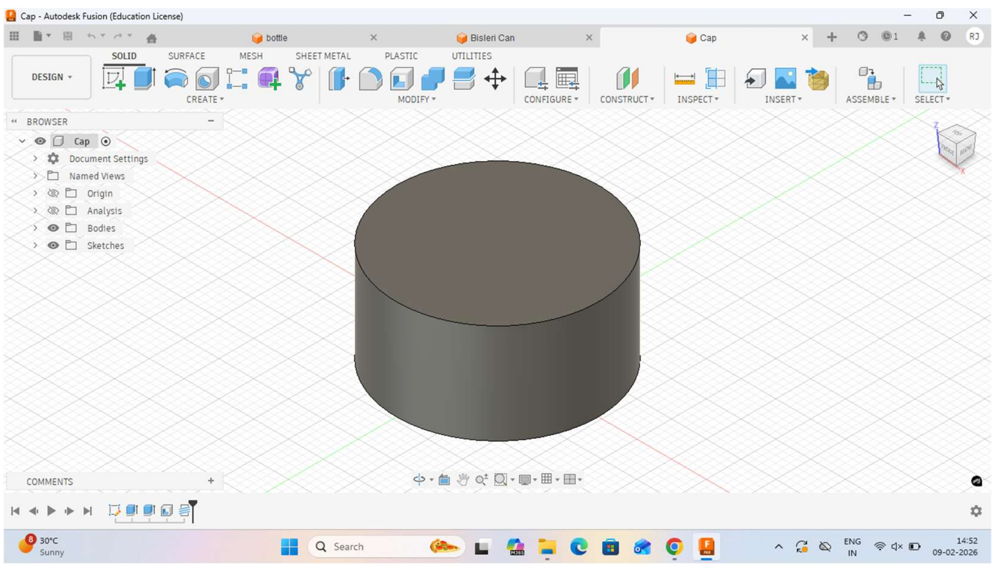
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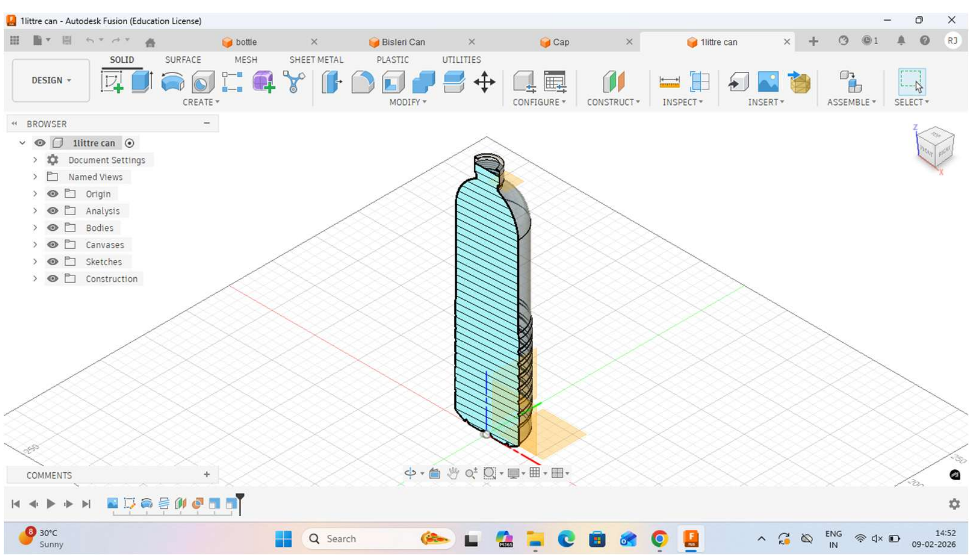
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4. Description of the Bottle Model

The bottle model was created in the **Solid workspace** of Autodesk Fusion 360. The design consists of a cylindrical body with a smooth shoulder and a threaded neck for the cap.

4.1 Bottle Body

The bottle body was designed by first creating a 2D sketch on the vertical plane. The sketch included the main cylindrical profile of the bottle. Using the **extrude feature**, the sketch was converted into a 3D solid body. Fillets were applied at necessary edges to smoothen transitions and avoid sharp corners.

4.2 Neck and Shoulder

The neck of the bottle was designed to support a screw cap. A smooth transition between the bottle body and neck was created using fillets, giving the bottle a realistic appearance. Threads were added to the neck region to allow proper engagement with the cap.

4.3 Text Embossing

Personalized text ("Roshan") was added to the surface of the bottle using the **text and emboss feature**. This demonstrates customization capabilities and branding possibilities in product design.

4.4 Appearance

A transparent red appearance was applied to the bottle to visually represent liquid storage and internal volume. This also improves the overall presentation and realism of the model.

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5. Description of the Cap Model

The cap was designed as a **separate component** to ensure modularity and ease of assembly.

5.1 Cap Geometry

The cap model consists of a cylindrical solid created using a circular sketch and extrusion. The dimensions were chosen carefully to match the threaded neck of the bottle.

5.2 Functional Design

The cap design ensures that it fits securely onto the bottle neck. Although the model is simple, it represents the base structure of a typical plastic bottle cap, which can be further enhanced with grip patterns or internal threads.

6. Assembly of Bottle and Cap

After completing the individual parts, the bottle and cap were assembled in the **Assembly workspace** of Autodesk Fusion 360.

6.1 Assembly Process

- The bottle body and cap components were inserted into the assembly environment.
- Proper alignment was ensured by positioning both components along the same central axis.
- Assembly constraints were applied to fix the cap on the bottle neck.

6.2 Final Assembly

The completed assembly accurately represents a real-world bottled water container. The alignment confirms that the cap fits correctly on the bottle without interference, validating the design dimensions.

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7. Methodology

The project was carried out using the following methodology:

1. Studying the basic structure of a commercial water bottle.
2. Creating sketches and converting them into 3D solids.
3. Applying solid modeling features such as extrude, fillet, and emboss.
4. Designing the cap as a separate part for flexibility.
5. Assembling the bottle and cap using assembly constraints.
6. Applying appearances to enhance visualization.
7. Reviewing the final model for accuracy and realism.

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8. Applications of the Design

- Packaging industry
- Bottled water and beverage manufacturing
- Product design and prototyping
- CAD training and educational purposes
- Custom bottle branding and visualization

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9. Conclusion

In this project, a **water bottle and cap were successfully designed and assembled using Autodesk Fusion 360**. The project provided hands-on experience in solid modeling, component creation, and assembly design. It demonstrated how CAD tools can be used to design everyday products with accuracy and efficiency.

The final model closely resembles a real commercial bottle and can be further enhanced by adding manufacturing details such as wall thickness analysis, stress testing, and realistic rendering. Overall, the project helped in developing essential CAD skills and understanding the complete product design workflow.